HACKTHEBOX

MEETUPS

Patras, Greece

Meetup #1

- Introduction.

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Wifi for Upatras.

Guest Wifi

https://my.upatras.gr/docs/sindesimotita/syndesi-sto-asyrmato-diktyo-upatras-guest/

Powerpoint Presentation.

CybersecurityWiki

github.com/nikolasfil/Cybersecurity-Wiki/wiki/meetup-01-introduction

Supported By

HackTheBox

ISSG

Information Systems Security Group, University of Patras

Agenda.

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- **Intros:** Introduction to Cybersecurity and CTFs.
- **Demos:** Showcasing CTF and Machines .
- **CSCG:** Cybersecurty Challenge Greece .
- **Hack-along:** Interactive Session on the dedicated labs.

- Intros.

Intro to Hacking & CTF.

Cybersecurity

Cybersecurity is the practice of protecting systems, networks, and data from digital attacks.

Temp Title

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aliquam euismod, metus in dapibus pulvinar, ipsum lacus gravida nisi, vel hendrerit libero nulla et risus. Praesent vel lacus eget risus cursus mattis. Etiam ut nibh nunc. Proin lobortis quis diam quis ultricies. Nullam rutrum arcu id erat dignissim, non finibus est luctus. Aenean vitae justo semper, bibendum justo congue, laoreet dui. Praesent vel lacus eget risus cursus mattis.

Intro to Cybersecurity .

Lorem ipsum dolor sit amet, consectetur adipiscing elit. In egestas auctor purus, ut commodo tortor auctor eu. **Cras suscipit lacus eu ex ullamcorper tincidunt.** Maecenas rhoncus elit ipsum, mollis tincidunt massa laoreet id. In egestas tellus a mollis molestie. Morbi euismod lacus nisl, ornare blandit urna faucibus id. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia curae; Suspendisse nec lacus tortor. Nam cursus euismod arcu eu aliquet. **Duis at interdum turpis.** Praesent venenatis nibh ac ultrices euismod. Curabitur luctus efficitur fermentum. Quisque venenatis vehicula sodales. Vestibulum placerat tellus rhoncus, egestas nisi in, posuere enim. **Sed varius.**

Lorem ipsum dolor sit amet, consectetur adipiscing elit. **Aliquam euismod, metus in dapibus pulvinar, ipsum lacus gravida nisi, vel hendrerit libero nulla et risus.** Praesent vel lacus eget risus cursus mattis. Etiam ut nibh nunc. Proin lobortis quis diam quis ultricies. Nullam rutrum arcu id erat dignissim, non finibus est luctus. Aenean vitae justo semper, bibendum justo congue, laoreet dui. Maecenas in odio ut tellus aliquam luctus sit amet ac sapien. **Aliquam erat volutpat. Etiam.**

Intro to Hacking.

Quotes

"Hacking is the art of exploiting weaknesses in a system to achieve a goal."
Jon Erickson, Hacking: The Art of Exploitation

"A hacker is someone who enjoys exploring the details of programmable systems and stretching their capabilities"
Charlie Miller, The Hacker's Handbook Series

Phases

- **Reconnaissance:** gathering information.
- **Initial Access - Execution:** initial foothold in the target system
- **Persistence:** maintaining access in a compromised system
- **Post Exploitation - Privilege Escalation:** obtaining admin rights
- **Discovery - Lateral Movement:** explore the compromised network

Intro to Capture The Flag.

Explanation

CTF stands for Capture The Flag, which is a type of cybersecurity competition where participants find and exploit vulnerabilities in systems to capture hidden "flags" or pieces of information. These competitions help develop and test cybersecurity skills in a gamified environment.

Categories

- **Cryptography:** Challenges involving encryption, hashing, and encoded data. The goal is usually to decode, decrypt, or analyze cryptographic systems to recover hidden information.
- **Web:** Focuses on vulnerabilities in web applications such as authentication flaws, SQL injection, XSS, and misconfigurations.
- **Reverse Engineering:** Involves analyzing compiled programs or binaries to understand how they work
- **Forensics:** Challenges based on analyzing digital artifacts such as network captures, disk images, logs, or memory dumps
- **Miscellaneous:** A broad category that includes puzzles or challenges that don’t fit into other categories
- **Pwn:** Challenges focused on exploiting vulnerabilities in compiled programs.

Useful Linux Commands.

Generic Commands

ls	list files
cd	change directory
pwd	show current location
cat	print file
less	view large files
cp	copy files
mv	move / rename
rm	delete
man	help

Useful in Hacking

grep	find secrets
find	locate sensitive files
ps	inspect processes
ss	discover open ports
curl	attack web services
chmod	change permissions
history	find credentials
python3	scripting & payloads
whoami	information about user

- Tools.

nmap.

Description

Network Mapper is a network scanning tool used to discover hosts, open ports, running services, and operating systems on a network..

Common Uses

- Discover Live hosts
- Identify open ports
- Detect services and versions
- OS fingerprinting.

```
 - Check if an IP address is up, and guess the remote hosts operating system:
   nmap -O ip_or_hostname

 - Try to determine whether the specified hosts are up (ping scan) and what their
names and MAC addresses are:
   sudo nmap -sn ip_or_hostname optional_another_address

 - Also enable scripts, service detection, OS fingerprinting and traceroute:
   nmap -A address_or_addresses

 - Scan a specific list of ports (use '-p-' for all ports from 1 to 65535):
   nmap -p port1,port2,...,portN address_or_addresses

 - Perform service and version detection of the top 1000 ports using default NSE
scripts; writing results ('-oN') to output file:
   nmap -sC -sV -oN top-1000-ports.txt address_or_addresses

 - Scan target(s) carefully using 'default and safe' NSE scripts:
   nmap --script "default and safe" address_or_addresses

 - Scan web server running on standard ports 80 and 443 using all available 'http-
*' NSE scripts:
   nmap --script "http-*" address_or_addresses -p 80,443

 - Perform a stealthy very slow scan ('-T0') trying to avoid detection by IDS/IPS
and use decoy ('-D') source IP addresses:
   nmap -T0 -D decoy1_ipaddress,decoy2_ipaddress,...,decoyN_ipaddress
```

gobuster.

Description

Gobuster is a tool for brute forcing directories, DNS subdomains, and virtual hosts written in go..

Common Uses

- directory enumeration
- dns subdomain enumeration
- fuzzing
- s3 aws bucket enumeration
- tftp enumeration
- vhost enumeration

```
- Discover directories and files that match in the wordlist:
  gobuster dir --url https://example.com/ --wordlist path/to/file

- Discover subdomains:
  gobuster dns --domain example.com --wordlist path/to/file

- Discover Amazon S3 buckets:
  gobuster s3 --wordlist path/to/file

- Discover other virtual hosts on the server:
  gobuster vhost --url https://example.com/ --wordlist path/to/file

- Fuzz the value of a parameter:
  gobuster fuzz --url https://example.com/?parameter=FUZZ --wordlist
path/to/file

- Fuzz the name of a parameter:
  gobuster fuzz --url https://example.com/?FUZZ=value --wordlist path/to/file
```

ffuf.

Description

ffuf is a fast web fuzzer written in Go that allows typical directory discovery, virtual host discovery (without DNS records) and GET and POST parameter fuzzing.

Common Uses

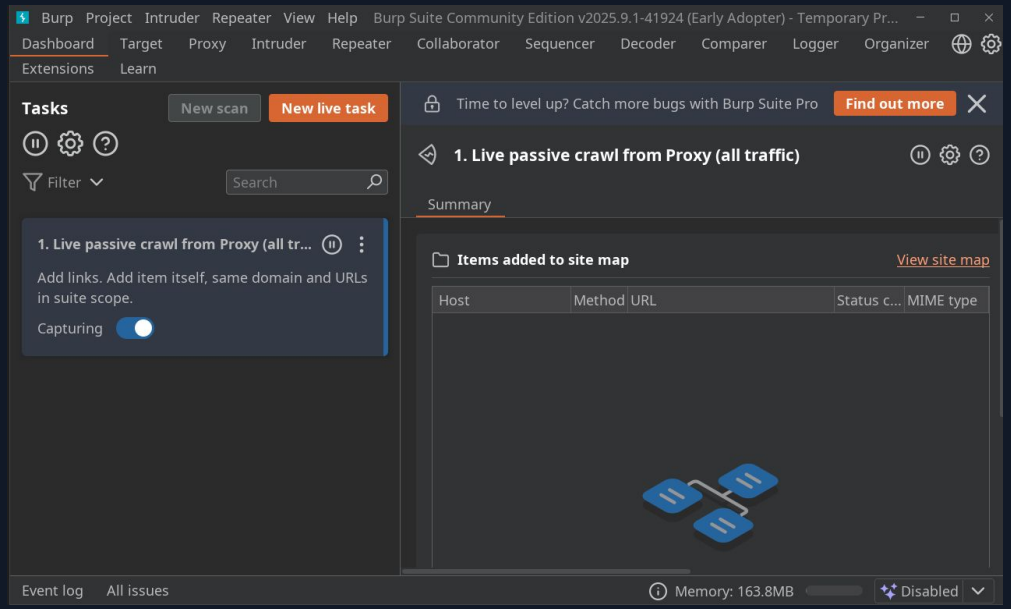
- directory enumeration
- dns subdomain enumeration
- fuzzing
- s3 aws bucket enumeration
- tftp enumeration
- vhost enumeration

```
  - Enumerate directories using [c]olored output and a [w]ordlist specifying a
target [u]RL:
    ffuf -c -w path/to/wordlist.txt -u https://example.com/FUZZ

  - Enumerate webservers of subdomains by changing the position of the keyword:
    ffuf -w path/to/subdomains.txt -u https://FUZZ.example.com

  - Fuzz with specified [t]hreads (default: 40) and pro[x]ying the traffic and
save [o]utput to a file:
    ffuf -o -w path/to/wordlist.txt -u https://example.com/FUZZ -t 500 -x http://12
7.0.0.1:8080

  - Fuzz a specific [H]eader ("Name: Value") and [m]atch HTTP status [c]odes:
    ffuf -w path/to/wordlist.txt -u https://example.com -H "Host: FUZZ" -mc 200

  - Fuzz with specified HTTP method and [d]ata, while [f]iltering out specific
status [c]odes and response [s]ize:
    ffuf -w path/to/postdata.txt -X POST -d "username=admin\&password=FUZZ" -u
https://example.com/login.php -fc 302,401-499 -fs 1234

  - Fuzz multiple positions with multiple wordlists using different modes and
[a]uto [c]alibration to reduce false positives:
    ffuf -w path/to/keys:KEY -w path/to/values:VALUE -mode pitchfork|clusterbomb -
u https://example.com/id?KEY=VALUE -ac

  - Proxy requests through a HTTP MITM pro[x]y (such as Burp Suite or mitmproxy):
    ffuf -w path/to/wordlist -x http://127.0.0.1:8080 -u https://example.com/FUZZ
```

burpsuite.

Description

Burp Suite is a web security testing platform used for intercepting, modifying, and analyzing HTTP traffic.



Common Uses

- Intercept web traffic
- Modify requests
- Test authentication flows
- Find vulnerabilities

sqlmap.

Description

sqlmap is an automated tool used to detect and exploit SQL injection vulnerabilities.

Common Uses

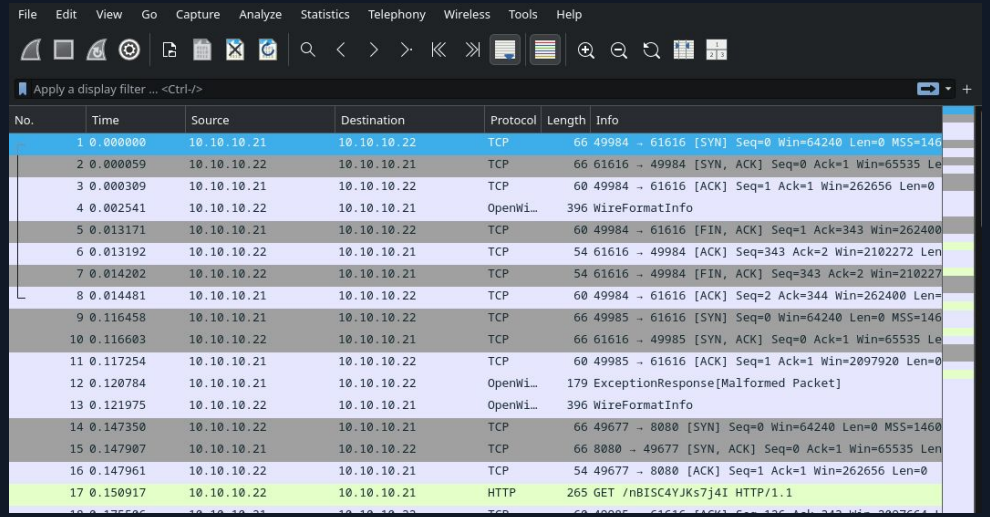
- Detect SQL injection
- Dump databases
- Extract credentials
- Automated exploitation

```
  Detect and exploit SQL injection flaws.
  More information: https://github.com/sqlmapproject/sqlmap/wiki/Usage.

  - Run sqlmap against a single target URL:
    python sqlmap.py --url "http://www.example.com/vuln.php?id=1"

  - Send data in a POST request (--data implies POST request):
    python sqlmap.py --url "http://www.example.com/vuln.php" --data="id=1"

  - Change the parameter delimiter (& is the default):
    python sqlmap.py --url "http://www.example.com/vuln.php" --
data="query=foobar;id=1" --param-del=";"

  - Select a random User-Agent from ./txt/user-agents.txt and use it:
    python sqlmap.py --url "http://www.example.com/vuln.php" --random-agent

  - Provide user credentials for HTTP protocol authentication:
    python sqlmap.py --url "http://www.example.com/vuln.php" --auth-type Basic --
auth-cred "testuser:testpass"
```

wireshark.

Description

wireshark is a network protocol analyzer used to inspect and analyze network traffic.



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.10.10.21	10.10.10.22	TCP	66	49984 → 61616 [SYN] Seq=0 Win=64240 Len=0 MSS=146
2	0.000059	10.10.10.22	10.10.10.21	TCP	66	61616 → 49984 [SYN, ACK] Seq=0 Ack=1 Win=65535 Le
3	0.000309	10.10.10.21	10.10.10.22	TCP	60	49984 → 61616 [ACK] Seq=1 Ack=1 Win=262656 Len=0
4	0.002541	10.10.10.22	10.10.10.21	OpenWi…	396	WireFormatInfo
5	0.013171	10.10.10.21	10.10.10.22	TCP	60	49984 → 61616 [FIN, ACK] Seq=1 Ack=343 Win=262400
6	0.013192	10.10.10.22	10.10.10.21	TCP	54	61616 → 49984 [ACK] Seq=343 Ack=2 Win=2102272 Len
7	0.014202	10.10.10.22	10.10.10.21	TCP	54	61616 → 49984 [FIN, ACK] Seq=343 Ack=2 Win=210227
8	0.014481	10.10.10.21	10.10.10.22	TCP	60	49984 → 61616 [ACK] Seq=2 Ack=344 Win=262400 Len=
9	0.116458	10.10.10.21	10.10.10.22	TCP	66	49985 → 61616 [SYN] Seq=0 Win=64240 Len=0 MSS=146
10	0.116603	10.10.10.22	10.10.10.21	TCP	66	61616 → 49985 [SYN, ACK] Seq=0 Ack=1 Win=65535 Le
11	0.117254	10.10.10.21	10.10.10.22	TCP	60	49985 → 61616 [ACK] Seq=1 Ack=1 Win=2097920 Len=0
12	0.120784	10.10.10.21	10.10.10.22	OpenWi…	179	ExceptionResponse[Malformed Packet]
13	0.121975	10.10.10.22	10.10.10.21	OpenWi…	396	WireFormatInfo
14	0.147350	10.10.10.22	10.10.10.21	TCP	66	49677 → 8080 [SYN] Seq=0 Win=64240 Len=0 MSS=1460
15	0.147907	10.10.10.21	10.10.10.22	TCP	66	8080 → 49677 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len
16	0.147961	10.10.10.22	10.10.10.21	TCP	54	49677 → 8080 [ACK] Seq=1 Ack=1 Win=262656 Len=0
17	0.150917	10.10.10.22	10.10.10.21	HTTP	265	GET /nBISC4YJKs7j4I HTTP/1.1

Common Uses

- Packet inspection
- Credential discovery
- Protocol analysis
- Traffic investigation

Other useful tools

- **cyberchef:** web-based tool used for data analysis, encoding, decoding, encryption and transformation
- **ghidra:** reverse engineering framework developed by the NSA, used to analyze compiled binaries
- **metasploit:** penetration testing platform used to develop, test and execute exploits against vulnerable systems
- **revshells.com:** site useful for building a reverse shell command
- **pwncat:** a more advanced reverse shell
- **linpeas:** privilege escalation auditing tool for linux systems
- **pwntools:** python library used in exploitation
- **gdb:** debugger, useful for reverse engineering challenges, pwn, binary
- **wordlists**: https://github.com/kkrypt0nn/wordlists
- **pentestmonkey: https://pentestmonkey.net**

- CTF.

CTF Challenges.

CryptOfTheUndead

Very Easy.
ReverseEngineering

El Pipo

Very Easy
Pwn

Data Siege

Medium
Forensics

Virtual Machines.

Easy Machines.

Build

easy

Oopsie

very easy

Soulmate

easy

CodePartTwo

easy

Broker

easy

Not Easy Machines.

Cache

medium

Ambassador

medium

Heal

medium

Developer

hard

Skyfall

insane

Carpediem

hard

- CSCG.

Greece Cybersecurity Challenge.

Info

13th of March, *University of Peloponnese*

Cyber security event for students in Greece. A scenario-based game played by teams of maximum 4

REGISTER NOW!

cybersecuritychallenge.gr

SANS ON-DEMAND COURSES

For each member of the winning team, **SANS Institute** gives out 1 On-Demand Course for any SANS course of your choice.

CISCO CERTIFIED NETWORK ASSOCIATE COURSE & LPI SECURITY ESSENTIALS CERTIFICATION VOUCHER

For each member of the second ranked team, **SCP School of Certified Professionals** offers a CCNA

SILVER ANNUAL ACADEMY & FORTINET ACCESS POINT

For each member of the third ranked team, **HackTheBox** gives out 1 Silver Annual Academy subscription and **GCC Hellas** gives out 1 Fortinet Access Point.

VIP+ ANNUAL LAB SUBSCRIPTION

For each member of the fourth ranked team, **HackTheBox** gives out 1 VIP+ Annual subscription.

3-MONTH PROLAB SUBSCRIPTION

For each member of the fifth ranked team, **HackTheBox** gives out a 3-month Prolab subscription

Looking for a team.

Google Form

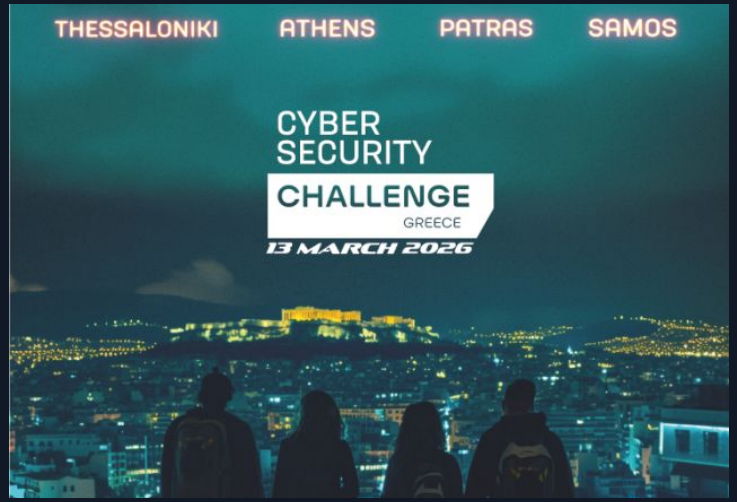
If you are looking for a team or are looking to add members in your group

https://forms.gle/BvviaPTRTR9TfjpL9

Greece Cybersecurity Challenge.

What is it

https://cybersecuritychallenge.gr/2025
Cyber security event for students in Greece. A scenario-based game played by teams of maximum 4



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Prizes

SANS ON-DEMAND COURSES	CISCO CERTIFIED NETWORK ASSOCIATE COURSE & LPI SECURITY ESSENTIALS CERTIFICATION VOUCHER	SILVER ANNUAL ACADEMY & FORTINET ACCESS POINT	VIP+ ANNUAL LAB SUBSCRIPTION	3-MONTH PROLAB SUBSCRIPTION
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CSCG Workshop.

Information

On the day of the competition at 10:00, the workshop “Industrial Security” will take place, by Chris Pliatsikas

- **Temp Title 1:** Vivamus sed tempor nunc, non congue velit. Morbi tempus vulputate orci, in pulvinar elit accumsan nec. Fusce augue erat, faucibus egestas.
- **Temp Title 2:** Aenean vitae justo semper, bibendum justo congue, laoreet dui. Maecenas in odio ut tellus aliquam luctus sit amet ac sapien. Aliquam erat volutpat. Etiam.
- **Temp Title 3:** Etiam ut nibh nunc. Proin lobortis quis diam quis ultricies. Nullam rutrum arcu id erat dignissim, non finibus est luctus.
- **Temp Title 4:** Nulla arcu turpis, ullamcorper scelerisque leo quis, dictum malesuada odio. Lorem ipsum dolor sit amet, consectetur adipiscing elit.

HACKTHEBOX

MEETUPS

Patras, Greece

Thank you!